

Module 5

Information Management

Answers

The suggested answers are longer than what candidates are expected to give in the examination. The purpose of the suggested answers is meant to help candidates in their revision and learning. The suggested answers may not contain all the correct points and candidates should note that credit will be awarded for valid answers which may not fully covered in the suggested answers.

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer 1(a)

Data and information may have the following characteristics related to quality:

- Effectiveness: Data and information must be correct, timely, consistent and relevant to the purpose for which it is being used, and must be delivered in a way that is usable.
- Integrity: Data and information must be accurate, complete and consistent.
- Reliability: Management must be able to depend on the data and information provided, as well as on its availability to fulfil their fiduciary and governance duties.
- Verifiability: The source and substance of raw data and information must be able to be confirmed or, when calculations or processing is involved, the results must be reproducible by a third party using the same data or information.

Answer 1(b)

Information systems have the following three roles:

- Information storage, processing, consolidation and accessibility: At its most basic level, an information system captures and stores data, and makes relevant data more accessible to all authorised personnel who need it.
- Resource planning and control: Management relies on information systems to provide information and insight into all organisational resources. Information systems help managers optimise the use of resources to meet the organisation's objectives.
- Decision making: When anyone in the organisation needs to make a business decision, that person should be able to consult the information system for information and insight. This can be valuable in identifying opportunities and deciding how to capitalise on those opportunities.

Answer 1(c)

The following types of e-commerce activities could be done by an online fashion store:

- online marketing and advertising;
- online sales;
- online customer service; and
- digital billing, payments and refunds.

Answer 1(d)(i)

Airline industry: Customers may book their own flights, choose their seats and meals as they want.

Answer 1(d)(ii)

Catering industry: Customers may order exactly what they want, choose pickup or delivery and even reserve a table without having to deal with awkward and confusing phone calls.

Answer 1(d)(iii)

Banking industry: Customers may pay bills, transfer money, check their balance and many other banking activities at any time of day or night.

Answer 2(a)(i)

When an organisation considers changing the type of information system, accountants must be involved in decisions, such as evaluating which software to purchase, how to design software or systems, and the implementation of software or systems. Accountants identify the financial data required, how the data flows into the system (input), how that information must be reported (output) and the accounting rules that must be applied. Accountants are also involved in the design of other information systems that may contain accounting information.

Answer 2(a)(ii)

Accountants are required to evaluate and judge whether the data and information in a particular information system are complete and accurate. Before having a conclusion and/or judgement, they should collect and evaluate evidence that is stored in or generated by the information system. They should also assess the effectiveness of the system's information technology (IT) controls and the organisation's degree of compliance with prescribed standards.

Answer 2(b)(i)

Human resources:

- Applicant records, such as applications and resumes, to select and invite appropriate applicants for interviews;
- List of positions within the organisation to understand basic job allocations among the staff;
- List of skills needed for the organisation to function to see whether there is a need to hire a new staff or transfer an existing staff internally;
- Staff benefits information to study why there is a high staff turnover rate; and
- Labour laws and policies to maintain compliance.

Answer 2(b)(ii)

Production:

- Product designs and specifications to facilitate product differentiation;
- Inventory records to understand the movements of products in a specific period;
- Employee schedules to know whether there is a need to hire part-time workers;
- Labour costs to see the impact of adopting new MPF rules; and
- Maintenance records and schedules to study why there is a high malfunction rate among the machines in the past two months.

Answer 2(c)(i)

Information systems can improve processes continuously by performing or facilitating the following tasks:

- Analysing processes to determine whether they are still necessary.
- Automating existing manual processes using expert systems (ESs).
- Redesigning or reengineering processes to make them more efficient.
- Expanding the data flows to reach more functions to make it possible for sequential activities to occur in parallel.
- Creating innovative business processes that, in turn, generate more business.

Answer 2(c)(ii)

Nowadays, business communications rely on networking infrastructure and communications tools such as e-mail, shared calendars or some tools in an information system as follows:

- Project management systems facilitate communication among team members by allowing them to set milestones, mark milestones as met and flag problems that are slowing progress.
- Shared calendars facilitate planning and the scheduling of meetings.
- Presence services indicate the location and availability of people in real time, making it easier to find and contact specific individuals.
- Collaboration tools enable people to work on projects together and often have integrated communication tools, such as notes, task lists and schedules that can be used to keep everyone who is involved in the project informed.
- Inventory systems can track when materials are checked out, and automatically order materials to replenish supplies. Managers can check the system to find out when the materials will arrive.

Answer 3(a)

The following reports can monitor the performance of a canned accounting software:

- Technical performance reports on measurements, such as storage, network, computer processor utilisation and other factors that can impact performance. Performance impacts include slow system response, failure of the system to respond to commands and system errors.
- Availability reports, including system downtime from problems and system maintenance.
- System usage reports, including number of users, peak usage times and number of transactions.

Answer 3(b)

Existing system review and meeting with users: System analysis
Layout customisation among reports and documents: System implementation
System training: System implementation

Answer 3(c)

To ensure that user needs have been met, many organisations require user acceptance at the end of the implementation of a new system.

When system testing is completed, the manager-in-charge and his/ her primary users of the system should discuss whether they are satisfied with the system functions, their performance as well as the testing results. If the feedback is positive, then the manager will sign an acceptance agreement on behalf of his/ her department or organisation.

The enforcement of user acceptance requires that project teams seek and consider users' feedback and that they work hard to meet the needs of users.

Answer 3(d)

The management of Williams may do the following in selecting a suitable ERP system: (Any three)

- Setup a committee to identify evaluation criteria among ERP systems;
- Identify vendors of ERP systems by web search or referral;
- Evaluate ERP systems according to pre-defined evaluation criteria; and
- Select an ERP vendor based on evaluation results.

Answer 4(a)

Artificial intelligence ("AI") is a subfield of computer science that focuses on developing machines that mimic the human ability to perform tasks that normally require human intelligence.

Answer 4(b)

Strong AI is a hypothetical artificial intelligence that matches or exceeds human intelligence and creativity. With strong AI, a computer would have consciousness and independent thought.

Weak AI is the ability to perform a useful and specific characteristically human function as well or better than humans.

Answer 4(c)

A performance dashboard has the following capabilities:

- drilling down;
- identifying critical success factors (CSFs);
- setting and monitoring key performance indicators (KPIs);
- conducting trend analysis; and
- generating exception reports.

Answer 4(d)(i)

A business intelligence system may use:

- a line chart or a bar chart to present the monthly, quarterly or even yearly net profit ratio. Users may know the movement of this ratio and recognise if there is an increase in the current period; or
- an arrow symbol to represent an increase of net profit ratio in the current period compared to the last period.

Answer 4(d)(ii)

A business intelligence system may use:

- red font colour to indicate there is a shortage of an inventory item; or
- a meter to show the current inventory level and indicate there is a shortage of an inventory item.

Answer 5(a)

Security policies should have the following components:

- Overview – This is generally a paragraph or two that explains why the organisation decided it would be appropriate to have the policy.
- Scope – The scope explains who or what is covered by the policy.
- Definitions – This optional section includes definitions of terms that may not be generally known to the policy's audience, such as technical terms.
- Statement – The policy statement formulates how the organisation will deal with a particular situation. In short, it explains the policy.
- Enforcement – This section specifies the consequences for violating a policy. It may refer to other policies (e.g., Human Resources policies) for penalties.

Answer 5(b)

We can classify the data of non-governmental organisations from IT security perspective or from Big Data perspective.

From IT security perspective, the data can be classified into the following three categories:

- Confidential Data: Confidential information that requires the highest level of security. Unauthorised release of this information is a significant regulatory or security risk.
- Restricted Data: Confidential information that requires a moderate level of security. Unauthorised release of this information is a medium regulatory or security risk.
- Internal Use Data: Information that may not be confidential by law, regulation or contract, but could have a negative impact if released without authorisation. This could include documents for internal organisation use, such as reports or internal communications.
- Public Data: Non-confidential data that is not subject to security control requirements. The release of public data poses little or no risk to an organisation.

From Big Data perspective, the data can be classified into the following categories:

- Traditional organisation data: Data and/ or information collected or processed from transactions, from point-of-sales (POS), enterprise resource planning (ERP) and/ or customer relationship management (CRM) systems
- Machine-generated/ sensor data: Data generated from smart meters; manufacturing sensors; sensors integrated into smartphones, automobiles, airplane engines and industrial machines; and equipment logs data.
- Social data: Such as customer feedback comments; posts on microblogging sites such as Twitter; and content and interactions on social media sites such as Facebook, YouTube and LinkedIn.
- Images and videos captured by billions of devices located throughout the world, from digital cameras and camera phones to medical scanners and security cameras.

Answer 5(c)

Passwords suffer from the following limitations:

- Long or complex passwords may be hard to remember, so people write them down where others can see.
- Passwords can be identified by hackers through various means, such as cracking software or brute force attacks.
- If only a password is used for authentication, it does not provide strong identity check. This is because a user may tell his/ her couple/ friend(s)/ colleague(s), who is not an authorised and legitimate user, about his/ her user name and password for accessing a system (or a computer/ an e-mail account) or ordering goods/ services on behalf of that user.

* * * END OF EXAMINATION PAPER * * *